

Universe Crystal Canonical Baseline

Official English Edition

Version 2.9

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Revision History

Version	Date	Revision Summary
V1.6		Initial Canonical Baseline establishing the overall philosophical framework of the Universe Crystal project.
V1.7		Clarified the ontological definition of Open Texture. Open Texture was formally defined as the Geometric Existence of an Organizational Interaction. The chapters on Time and Canonical Conclusions were updated accordingly, while the First Principles, Organization Hierarchy, and Dynamics remained unchanged.
V1.8		Introduced the Universe Crystal Grand View and formally established the Organizational Interpretation Principle, forming the three-layer theoretical structure of Framework – Ontology – Epistemology. The First Principles, Organization Hierarchy, Dynamics, and existing Ontology remained unchanged.
V2.0		Established the Texture Evolution framework, introducing Open Texture, Self-Closed Texture, and Interlocking Texture; unified Texture as the medium of Relationship Propagation; established First-Level and Second-Level Organization; and introduced Organizational Degrees of Freedom and Organizational Space.
V2.1		Major conceptual consolidation following Research Notes 018 and 019. Developed Organizational Degrees of Freedom into Winding Modes, Organizational Accessibility, Organizational Reading, and Organizational Coupling. Established distinct organizational interpretations for electric charge and spin. Extended the framework to Color Organization through an independent internal two-dimensional weaving space. Established Interlocking Texture as an active carrier and coordinator of color charge, and introduced the preliminary $9 \rightarrow 8$ color coordination structure and continuous three-body closure interpretation of color confinement.
V2.2		Corrected an internal semantic inconsistency in the classification of the Color Organizational Degree of Freedom. Removed the statement that classified Color as an internally organized Hidden Organizational Degree of Freedom of the First-Level Stable Closed Texture. No new theoretical interpretation was introduced in this revision.
V2.3		Added Research Note 021 to Part I as an appendable canonical research extension. Clarified that Color Charge is not an intrinsic organizational property of an isolated First-Level Organization, but emerges as a relational phase when First-Level Organizations participate in Second-Level Weaving; clarified the role of Interlocking Texture in carrying and coordinating color-phase change relations; and clarified the organizational motivation for the proposed two-degree-of-freedom color structure. Part II remains reference, summary, and status material.
V2.4		Consolidated the Canonical Baseline into a stable chapter architecture. Existing topics retain one canonical chapter each; duplicate chapters are removed; Research Note 021 is integrated into the existing Color chapters rather than added as a duplicate topic chapter; Time and Space are established as Chapters 19 and

		20; and Part II adopts an independent R-series reference numbering system. A permanent Canonical Chapter Stability Rule is established so future topic updates modify existing chapters while genuinely new topics receive new chapters without renumbering or reordering existing chapters.
v2.5	Aug 15 2026	Refined Organizational Reading through the provisional concept of multi-cycle organizational closure and finite-cycle reading, and extended the Spin chapter with the exploratory hypothesis that spin-1/2 may correspond to a two-cycle organizational closure structure. Quantum Superposition is treated as an interpretation arising from the existing Spin and Organizational Reading framework rather than as a new canonical topic. RN 023
V2.6	Aug 15 2026	Incorporated the research conclusions of Research Note 024 into Chapter 12, establishing phase completion as the organizational basis of charge magnitude, winding direction as the organizational basis of charge sign, relative winding as the basis of attraction and repulsion, and continuing interaction through Open Texture as the basis of electric force.
V2.7	Aug 15 2026	Incorporated the research conclusions of Research Note 026 into Chapter 18, defining Mass as the physical manifestation of Organizational Magnitude and establishing its hierarchical structure as inherited organizational magnitude plus newly formed organizational magnitude at higher levels.
V2.8	Aug 15 2026	Incorporated the research results of Research Note 027. The Spin chapter was updated with the two-cycle organizational closure interpretation of spin-1/2. A new Entanglement chapter was added, establishing coupling of propagation closures as the organizational interpretation of entanglement and incorporating two candidate mechanisms: coupling of closure shapes and propagation closure across two particles.
V2.9	Aug 15 2026	Incorporated the research conclusions of Research Note 025 into Chapter 20 — Space. Space is interpreted as an emergent organizational property arising from Collective Closure, and the three-member minimum Collective Closure is proposed as the structural basis for three spatial degrees of freedom and three-dimensional space.

1. First Principles

1.1 Open Texture

Open Texture is the most fundamental organizational texture generated through interactions between organizations.

It is not matter.

It is not energy.

It is not a field.

Nor is it a trajectory through space.

Open Texture is the Geometric Existence of an Organizational Interaction.

Every organizational interaction gives rise to an Open Texture, which permanently becomes part of the organizational structure of the Universe Crystal.

Therefore, every Open Texture belongs to the Universe Crystal as a whole.

There is no such state as an interaction that is “half propagated.”

Open Texture constitutes the Fundamental Organizational Texture of the Universe Crystal.

1.2 Stable Closed Texture

A Stable Closed Texture is formed when an Open Texture achieves Stable Closure.

This gives rise to the First-Level Stable Closed Texture.

Within the current framework, electrons and quarks are regarded as research objects belonging to the First-Level Stable Closed Texture.

Closure is neither a closure in space nor a closure in time.

It is a Stable Closure within the Whole Crystal across all space-time.

1.3 Organizational Degree of Freedom

An Organizational Degree of Freedom is not a classical spatial dimension.

It is defined as a fundamental organizational degree through which the Fundamental Organizational Texture forms different stable organizational structures.

An Organizational Degree of Freedom is an independent organizational channel through which Relationship Propagation forms and maintains organization.

Different Organizational Degrees of Freedom may support different stable Winding Modes.

At present, the following fundamental properties are proposed to originate primarily from Organizational Degrees of Freedom:

- Spin
- Electric Charge
- Color

Mass remains a holistic organizational property established after Stable Closure.

1.4 Organization Hierarchy

The universe evolves through the continuous emergence of higher levels of organization.

Its evolution is not characterized by the continuous creation of new fundamental particles.

2. Texture Evolution

2.1 Open Texture

Open Texture is the most fundamental Texture Pattern.

An Open Texture consists of two open ends, representing one complete Open Event.

Open Texture is capable of propagating relationships.

However, since relationship propagation has not yet formed a closure, no stable organization can emerge.

2.2 Self-Closed Texture

When the relationship propagation path of an Open Texture continuously returns to itself, a Self-Closed Texture is formed.

The relationship propagation path completes a Self Closure, eliminating the open ends and establishing a continuous and stable propagation loop.

This continuous closure gives the organization a persistent Identity and forms the First-Level Organization.

Therefore:

Self-Closed Texture forms the First-Level Organization through the self-closure of the relationship propagation path.

2.3 Interlocking Texture

When multiple First-Level Organizations further organize into a higher-level organization, a new Texture Pattern emerges: Interlocking Texture.

Each First-Level Organization remains complete and occupies an independent Organizational Degree of Freedom.

Within each First-Level Organization, Identity continues to be maintained by its own Self-Closed Texture.

Interlocking Texture coordinates the continuous propagation of relationships among multiple First-Level Organizations, allowing relationship propagation to circulate across multiple Organizational Degrees of Freedom.

As relationship propagation continues, the overall propagation path forms a Collective Closure.

The Collective Closure locks a Stable Organizational Topology, giving rise to a new Second-Level Organization.

Interlocking Texture therefore has two related organizational roles.

First, it coordinates Relationship Propagation across multiple Organizational Degrees of Freedom.

Second, in color organization, it actively carries and coordinates net color charge between the First-Level Organizations participating in the joint closure.

Therefore:

Interlocking Texture forms the Second-Level Organization by coordinating the Collective Closure of relationship propagation among multiple First-Level Organizations.

2.4 Organizational Degree of Freedom

Every First-Level Organization corresponds to an independent Organizational Degree of Freedom.

An Organizational Degree of Freedom represents the independent organizational dimension through which a First-Level Organization maintains its continuous organizational activity.

It is not a coordinate in geometric space.

During the formation of a Second-Level Organization:

- Every First-Level Organization remains complete.
- Every First-Level Organization occupies an independent Organizational Degree of Freedom.
- Every First-Level Organization maintains its own Self-Closed Texture.
- Interlocking Texture coordinates the continuous propagation of relationships among multiple Organizational Degrees of Freedom.
- The relationship propagation paths across multiple Organizational Degrees of Freedom eventually form a Collective Closure.

Therefore:

A Second-Level Organization is not established through direct connections between First-Level Organizations.

Instead, it emerges when relationship propagation across multiple Organizational Degrees of Freedom forms a Collective Closure.

2.5 Organizational Space

When relationship propagation across multiple Organizational Degrees of Freedom completes a Collective Closure, a stable Organizational Topology emerges.

This topology gives rise to the first continuous Organizational Domain.

An Organizational Domain possesses:

- Interior
- Boundary
- Exterior

These properties are not pre-existing geometric space.

Instead, they naturally emerge from the Collective Closure of relationship propagation.

Therefore:

Space is not a prerequisite for organization.

It is an Emergent Organizational Property arising from the Collective Closure of relationship propagation.

2.6 Texture Evolution Sequence

Open Texture

↓

Self-Closed Texture

↓

Interlocking Texture

The corresponding Organizational Evolution is:

- Open Texture → Open Event
- Self-Closed Texture → First-Level Organization
- Interlocking Texture → Second-Level Organization → Organizational Space

3. Organization Hierarchy

3.1 Level 0

Open Texture

3.2 Level 1

First-Level Stable Closed Texture

Examples:

- Electron
- Quark

Characteristics:

- Point-like particles
- Stable Closure
- No volume
- Independent Identity
- Independent Organizational Degrees of Freedom

3.3 Level 2

Second-Level Organization

Examples:

- Proton
- Neutron

Characteristics:

- Stable organizations formed by multiple First-Level Stable Closed Textures
- The first emergence of physical volume
- The first emergence of higher-level organizational Identity
- The first emergence of spatial degrees of freedom
- Collective organization through Interlocking Texture

3.4 Level 3

Third-Level Organization

Example:

- Atom

Characteristics:

- Formation of chemical elements
- Emergence of chemical properties
- Organizational level at which chemical reactions occur

3.5 Level 4 and Above

Examples include:

- Molecules

- Life
- Consciousness
- Society
- Higher organizational systems

These represent the continuous upward emergence of organizational complexity.

4. Electron Baseline

4.1 Stability

Originates from Stable Closure.

4.2 Point-like Nature

Represents the local manifestation of a globally stable closure.

4.3 Spin

Originates from an Organizational Degree of Freedom whose internal organization couples to emergent spatial organization.

4.4 Electric Charge

Originates from a Charge Organizational Degree of Freedom.

Its observed value is interpreted through Organizational Reading of the phase accumulated during an Organizational Reading Cycle.

4.5 Mass

Represents the holistic property established after Stable Closure is achieved.

All electrons are identical because their mode of Stable Closure is unique.

5. Organizational Laws

5.1 First-Level Closure

First-Level Closure establishes Existence.

Stable Closure gives rise to stable Identity.

Identity comes from Stable Closure.

5.2 Second-Level Organization

Second-Level Organization establishes organizational stability through Collective Closure and Stable Organizational Topology.

5.3 Third-Level Organization

Third-Level Organization establishes chemical elements and chemical properties.

5.4 Higher Organizational Levels

Higher organizational levels continue to emerge through organization.

The evolution of the universe follows a common principle:

Stability is achieved when higher-level organizations form new Stable Closures.

5.5 Organizational Propagation

Organization is maintained through Relationship Propagation.

5.6 No Static Existence

Organization cannot be reduced to a purely static state.

This principle applies directly to the interpretation of Winding Modes and color coordination.

5.7 Organizational Accessibility

Different Organizational Degrees of Freedom become observable through different organizational mechanisms.

5.8 Organizational Reading

Observable properties arise through the organizational process by which one organization reads another organizational structure or interacts with it.

5.9 Organizational Coupling

Different Organizational Degrees of Freedom may become observable through their coupling to other organizational structures.

Spin provides the current primary example.

5.10 Continuous Higher-Level Organization

The universe continuously gives rise to higher levels of organization.

6. Crystal Evolution

6.1

The universe underwent a fundamental phase transition.

During this transition, First-Level Stable Closed Textures, including electrons and quarks, were formed.

Afterwards, no new First-Level Stable Closed Textures are created.

Subsequent cosmic evolution is therefore characterized primarily by the continuous enrichment of higher levels of organization.

The Universe Crystal never overwrites its own history.

Previously formed textures are never removed.

The Universe Crystal can only continue to grow.

7. Dynamics

7.1 Chemical Reactions

Chemical reactions occur at the level of Third-Level Organization.

Their essence is the reorganization of Third-Level Organizations.

Throughout this process:

- First-Level Stable Closed Textures remain unchanged.
- Second-Level Organizations remain unchanged.

A chemical bond is not a new First Principle.

Rather, it is an organizational structure through which Third-Level Organizations jointly organize First-Level Stable Closed Textures.

7.2 Nuclear Reactions

Nuclear reactions occur at the level of Second-Level Organization.

Fusion is the reorganization of Second-Level Organizations.

Fission is likewise the reorganization of Second-Level Organizations.

Throughout both processes, First-Level Stable Closed Textures remain unchanged.

7.3 Color Dynamics

Color dynamics occur within the organizational structure of First-Level Organizations participating in Interlocking Texture.

Color transformation is not treated as an isolated property change.

It is a relational organizational process coordinated through Interlocking Texture.

When one First-Level Organization undergoes a color transformation, the corresponding net color is carried through Interlocking Texture and coordinated with another First-Level Organization within the same collective closure.

The resulting three-body organization therefore maintains its closure through continuous mutual weaving.

7.4 Strong-Interaction Scale Dependence

The present framework does not yet contain a quantitative organizational scale variable capable of deriving the running strength of the strong interaction.

Therefore, asymptotic freedom and confinement are currently separated into:

- structural interpretation;
- qualitative analogy;
- future quantitative research.

7.5 Current Dynamics Boundary

The Universe Crystal framework interprets dynamics through Relationship Propagation, Winding Modes, and organizational transformation.

The dynamics of color organization are described in Chapters 15–17.

The framework does not yet provide quantitative dynamical equations for the strong interaction.

8. Organizational Interpretation Principle

8.1 Core Definition

Every organization can form its own organizational state only on the basis of the propagation relations within the organizational system to which it belongs.

Based on these organizational relations, it produces its own interpretation of the universe.

Therefore, every output generated by an organization is, in essence, an Interpretation formed according to the propagation relations within its organizational system.

It is not the Universe Crystal itself.

8.2 Corollary 1 — Organizational Interpretation

Observation is not an ability unique to consciousness.

Any organization capable of forming an organizational state and producing an output performs Organizational Interpretation.

This includes, but is not limited to:

- Particles
- Atoms

- Molecules
- Crystals
- Biological organizations
- Scientific instruments
- AI systems
- Consciousness systems
- Scientific communities

The only difference lies in their respective levels within the Organization Hierarchy.

8.3 Corollary 2 — Experimental Measurement

Experimental measurement is a form of Organizational Interpretation.

The outputs of scientific instruments, including voltage, electric current, images, spectra, and experimental data, are interpretations generated according to the organizational structure of the measuring instrument.

8.4 Corollary 3 — Scientific Theory

Scientific theories are forms of Organizational Interpretation.

Through the continuous accumulation of propagation relations, the scientific community continually develops new interpretations of the universe.

Therefore, scientific theories are not the Universe Crystal itself.

They are organizational interpretations constructed by the scientific community.

8.5 Corollary 4 — Consciousness

Consciousness is also a form of Organizational Interpretation.

Based on the propagation relations within its own organizational system, consciousness constructs experience, memory, understanding of the world, understanding of time, and understanding of meaning.

Therefore, every understanding of the universe produced by consciousness remains an organizational interpretation.

9. Organizational Reading

9.1

Every measurement is an act of Organizational Reading.

Reading is not exclusive to consciousness.

Every measuring device acquires information by progressively reading organizational relations.

Organizational Reading is the fundamental process through which an organization constructs an interpretation.

It serves as the bridge through which the classical world emerges.

Organizational Reading may access different Organizational Degrees of Freedom through different mechanisms.

For example:

- Charge is accessed through accumulated phase.
- Spin is accessed through coupling to emergent spatial organization.
- Color is accessed through collective organization and Interlocking Texture.

9.2 Organizational Reading Cycle

For certain Organizational Degrees of Freedom, the observable property is produced through a complete Organizational Reading Cycle.

The internal organizational process itself remains inaccessible to direct reconstruction.

Instead, Organizational Reading records an accumulated organizational quantity generated during the reading cycle.

Electric charge is currently interpreted through this mechanism.

Thus:

Observed Electric Charge = Phase accumulated during one Organizational Reading Cycle.

This mechanism is distinct from the organizational reading of Spin.

9.3 Multi-Cycle Organizational Reading

An organization may contain a closure structure extending across multiple organizational cycles. A measurement interaction may access only a finite portion of this complete closure structure.

Therefore, the observed manifestation does not necessarily reconstruct the complete organizational closure.

Multi-Cycle Organizational Closure → Finite-Cycle Reading → Phase Manifestation

This provides a basis for interpreting cases in which a complete organizational structure produces discrete measurement manifestations through a limited reading interaction.

This interpretation remains exploratory and requires future mathematical formalization.

10. Organizational Accessibility

10.1 Hidden Organizational Degrees of Freedom

A Hidden Organizational Degree of Freedom remains internal to the First-Level Stable Closed Texture.

External organizations cannot reconstruct its complete internal organization.

Organizational Reading observes only its external manifestation.

The Charge Organizational Degree of Freedom belongs to this category.

10.2 Coupled Organizational Degrees of Freedom

A Coupled Organizational Degree of Freedom remains internally organized but directly couples to emergent Organizational Space.

Its complete internal organization remains hidden.

Measurement instead observes its spatial projection and response through organizational coupling.

The Spin Organizational Degree of Freedom belongs to this category.

10.3 Emergent Organizational Degrees of Freedom

Some Organizational Degrees of Freedom emerge only after higher-level organization.

Spatial Degrees of Freedom are an example.

They become directly accessible after Second-Level Organization establishes a stable Organizational Space.

11. Winding Mode

11.1

A Winding Mode describes the stable winding behavior of Relationship Propagation within a particular Organizational Degree of Freedom.

It is not a new Texture Pattern.

Instead:

Texture Pattern describes how organization is formed.

Winding Mode describes how Relationship Propagation is maintained within an already established organization.

Different Organizational Degrees of Freedom may therefore possess different Winding Modes.

Observable quantum properties can arise through different mechanisms of organizational reading and organizational coupling.

Winding Mode provides the internal organizational layer connecting Organizational Degrees of Freedom to observable quantum properties.

12. Charge Organizational Degree of Freedom

12.1

Charge is interpreted as a Winding Mode within the Charge Organizational Degree of Freedom.

The complete internal winding structure remains hidden from direct observation. Organizational Reading does not reconstruct the complete winding structure. Instead, measurement observes the phase associated with the organizational weaving through an Organizational Reading Cycle.

The magnitude of electric charge is associated with the degree of phase completion within the self-closing weaving pattern.

Phase Completion → Charge Magnitude

The sign of electric charge is associated with the winding direction or handedness of the organizational weaving.

Winding Direction → Charge Sign

Opposite winding orientations correspond to opposite charge signs within the present organizational interpretation. Organizational winding direction is distinct from ordinary spatial handedness.

Charged organizations remain embedded within Open Texture after self-closure. Their interaction is determined by their relative winding organization.

Same Winding Direction → Organizational Incompatibility → Separation → Repulsion

Opposite Winding Direction → Organizational Compatibility → Convergence → Attraction

Attraction and repulsion are therefore interpreted as different organizational manifestations of the interaction between differently wound charged organizations through Open Texture.

Electric force is interpreted as the observable expression of the continuing organizational interaction between charged organizations through Open Texture.

The observed charge values of elementary organizations remain expressed through their organizational phase. The electron is associated with one complete charge phase, while the fractional charge values of quarks remain associated with their respective organizational phase structures.

Composite charge arises from the organizational combination of the charge contributions of constituent organizations.

A complete mathematical derivation of the winding structure, the relationship between phase completion and observed charge values, the quantitative magnitude and distance dependence of electric force, and the electromagnetic field remains open research.

13. Spin Organizational Degree of Freedom

13.1 Spin, Closure Phase, and Coherence

Spin should not be interpreted using the same observational mechanism as electric charge.

Unlike Charge, Spin couples to the spatial degrees of freedom that emerge through higher-level organization.

Measurements therefore reveal quantized spatial responses rather than merely an accumulated phase.

Within the present framework:

- the internal Spin Organizational Degree of Freedom remains organizationally hidden;
- its organization couples to emergent spatial organization;
- Organizational Reading observes a quantized spatial response.

A persistent organization has a propagation closure and an associated closure phase.

Spin-1/2 may be associated with a two-cycle organizational closure structure, while a measurement interaction may access the organization through a single reading cycle.

This provides a possible organizational interpretation of the two complementary measurement manifestations associated with spin-1/2.

This interpretation remains an exploratory hypothesis and does not yet constitute a mathematical derivation of spin-1/2.

13.2 Spin and Two-Cycle Closure

Spin-1/2 may be associated with a two-cycle organizational closure structure.

Under this hypothesis, the complete organizational closure of the Spin Degree of Freedom requires two organizational cycles.

A measurement interaction may access the organization through a single reading cycle.

Therefore:

Two-Cycle Organizational Closure → Single-Cycle Reading → Phase Manifestation

The two measurement manifestations of spin-1/2 may therefore correspond to two complementary phase manifestations of the underlying two-cycle organizational closure.

Quantum superposition is provisionally interpreted as a measurement-language representation of a multi-cycle organizational structure under finite-cycle Organizational Reading, rather than as an independent Organizational Degree of Freedom.

This remains an exploratory hypothesis and does not yet constitute a mathematical derivation of spin-1/2.

14. Color Organizational Degree of Freedom

14.1

Color is interpreted as a distinct Organizational Degree of Freedom.

Color is independent of ordinary spatial orientation.

A spatial reorientation of a First-Level Organization does not transform its color state.

The current framework proposes an internal two-dimensional color weaving space.

Three color states can be represented as three symmetric states within this two-dimensional organizational space.

This internal color weaving space is distinct from ordinary physical space.

14.2 Organizational Level of Color Charge

A First-Level Organization is formed through the weaving and closure of Open Texture.

Color charge is associated with the next organizational process:

First-Level Organizations → Second-Level Weaving → Second-Level Organization

The objects participating in this second weaving process are already organized structures.

Color charge is therefore interpreted as a relational state arising within the higher-order weaving relationship among these First-Level Organizations.

Color Charge is not an intrinsic organizational property of an isolated First-Level Organization. It emerges as a relational phase when First-Level Organizations participate in Second-Level Weaving.

14.3 Relation to First-Level Organization

The First-Level Organization has its own internal organization and closure.

When it participates in Second-Level Weaving, its color state is defined within the higher-order organizational relationship in which it participates.

Color therefore belongs to the relational organization established through Second-Level Weaving.

14.4 Why the Higher-Order Phase Can Require Two Degrees of Freedom

Open Texture, before First-Level Organization exists, has no mass concept and no time concept within the Universe Crystal organizational framework.

The original weaving process therefore acts directly on the fundamental organizational texture.

Once a First-Level Organization exists, it possesses mass, and the emergence of organized persistence provides the conditions under which temporal description becomes meaningful.

Second-Level Weaving therefore acts between already organized entities possessing mass and temporal organization.

The phase of this higher-order weaving can consequently contain richer information than the phase of the original Open Texture being woven into a First-Level Organization.

A higher-order weaving phase may therefore require multiple organizational degrees of freedom.

The two-degree-of-freedom structure proposed for color is associated with this richer higher-order organizational state.

14.5 Color and Anticolor

A quark participates in Second-Level Weaving with a color state.

An antiquark participates with the corresponding anticolor state.

Color and anticolor are interpreted as corresponding conjugate orientations of the higher-order weaving phase.

The precise mathematical relationship between this organizational interpretation and the fundamental and anti-fundamental color representations remains future research.

14.6 Unified Organizational Picture

Open Texture → First-Level Weaving → First-Level Organization

Twisting phase → Electric Charge

Organized weaving pattern and quantum state → Spin

Then:

First-Level Organizations → Second-Level Weaving

Relational weaving phase → Color Charge

Color-phase change relations → Interlocking Texture

Continuous coordination → Collective Closure

This establishes a common organizational language for electric charge, spin, and color while preserving their distinct organizational levels and mechanisms.

The detailed organizational meaning of this space is developed in Chapters 14–16.

15. Color Coordination Through Interlocking Texture

15.1

Interlocking Texture is not a neutral or passive connector in color organization.

It actively participates in the coordination of color charge among the First-Level Organizations forming a collective closure.

The color structure therefore involves both:

- the color states expressed by First-Level Organizations participating in the collective organization;
- the Interlocking Texture through which their color organization is coordinated.

15.2 Color–Anticolor Candidate Space

Three colors and three corresponding anti-colors generate nine candidate color–anticolor coordination modes.

These nine combinations form the candidate space for color coordination through Interlocking Texture.

15.3 Exclusion of the Identity Mode

Among the nine candidate combinations, one special symmetric combination acts as an identity transformation.

Its overall effect returns any color state to itself.

It produces no actual color transformation.

The Universe Crystal framework already establishes the principle that there is no static existence.

Applied to color coordination, the identity operation therefore does not constitute an independent non-static weaving relation.

15.4 Eight Non-Static Color-Phase Change Relations

The resulting structure is:

$$3 \times 3 - 1 = 8$$

The present framework identifies eight non-trivial color-phase change relations.

15.5 Interlocking Texture as the Carrier of Phase-Change Relations

When the color phase of one First-Level Organization changes, the corresponding change remains coordinated with the other First-Level Organizations participating in the same Second-Level closure.

Therefore:

Interlocking Texture carries and coordinates the relations between color-phase changes of First-Level Organizations.

The organizational sequence is:

Color-Phase Change → Interlocking Texture → Color-Phase Change Relation → Coordination with another First-Level Organization → Collective Closure.

15.6 Eight Non-Trivial Color-Phase Change Relations

The existing three-color / three-anticolor candidate structure gives nine candidate combinations.

One symmetric combination functions as an identity transformation and produces no color-phase change.

Under the No Static Existence principle, it does not constitute an active color-change relation.

The remaining structure therefore contains:

$$3 \times 3 - 1 = 8$$

non-trivial color-phase change relations.

The canonical interpretation is:

Interlocking Texture carries and coordinates eight non-trivial color-phase change relations within Second-Level Weaving.

The formal mathematical representation of these relations and their correspondence with the eight gluon degrees of freedom remain open research topics.

The significance of these eight relations is the active set of color changes that Interlocking Texture must carry and coordinate within Second-Level Weaving.

The eight-mode result remains a structural interpretation. A complete mathematical representation of the candidate operations and a formal correspondence with the eight gluon degrees of freedom remain future research.

16. Three-Body Joint Closure and Color Confinement

16.1

Interlocking Texture is not a container into which three already completed First-Level Organizations are placed.

It is the organizational constraint through which three First-Level Organizations maintain joint closure.

Color transformation is therefore part of the mechanism that continuously maintains the collective organization.

16.2 Color Transformation as a Relational Event

When one First-Level Organization undergoes a color transformation, the corresponding color-phase change relation is carried through Interlocking Texture and coordinated with another First-Level Organization within the same collective closure.

No color transformation is treated as an isolated organizational event.

Each transformation participates in the relational organization of the complete three-body structure.

16.3 Continuous Joint Closure

The three First-Level Organizations continuously participate in a mutual process of:

weaving → transformation → color-phase change → renewed weaving → renewed closure

Closure is therefore continuously maintained through internal transformation.

A proton is consequently interpreted, within the present framework, as three First-Level Organizations participating in a continuously maintained mutual weaving structure.

16.4 Structural Interpretation of Color Confinement

An isolated colored First-Level Organization cannot sustain the same joint closure mechanism outside the continuously maintained interlocking process.

Color confinement is therefore interpreted as the inability of an isolated colored organization to maintain the collective closure through which the color organization is continuously sustained.

This provides a structural organizational interpretation of confinement.

16.5 Color Closure

The three First-Level Organizations forming a proton participate in a continuously maintained Second-Level Weaving process.

Their color phases remain mutually coordinated through Interlocking Texture.

Color closure is therefore the continuous coordination of the color phases of the participating First-Level Organizations within collective closure.

The proton is consequently interpreted as a Second-Level Organization maintained through continuous mutual weaving.

It does not yet constitute a quantitative derivation of confinement or hadronization.

17. Asymptotic Freedom and Confinement: Current Boundary

17.1

The Universe Crystal framework currently distinguishes between structural interpretation and qualitative analogy.

The present research establishes a structural interpretation of:

- color organization;
- color–anticolor coordination;
- eight non-trivial color-phase change relations;
- continuous three-body joint closure;
- the organizational basis of color confinement.

However, the framework does not yet contain a quantitative variable describing how the strength of the strong interaction changes with scale.

The familiar directional behavior is:

- weaker interaction at short distance;
- stronger interaction at long distance;
- eventual hadronization rather than liberation of isolated color.

The current Universe Crystal interpretation uses an interlocking-ring picture as a qualitative analogy.

Three interlocking structures may be imagined as becoming relatively relaxed when close together and increasingly strained as they are pulled apart.

This picture conveys the direction of:

short-distance freedom → long-distance strengthening → confinement / hadronization

However, it does not derive the precise rate of scale dependence.

Therefore:

The ring-tension picture is a qualitative analogy, not a quantitative derivation.

A formal organizational scale variable remains an open research problem.

18. Mass

18.1 Mass as Organizational Magnitude

Mass is interpreted as the physical manifestation of Organizational Magnitude incorporated into a stable organization.

Organizational Magnitude describes the amount of organized relational structure incorporated into an organization. It is distinct from spatial extension and does not require pre-existing spatial volume.

Organizational Magnitude is provisionally associated with two complementary properties:

Organizational Multiplicity — the number of distinguishable relationship-propagation instances incorporated into the organization.

Weaving Density — the amount of organizational structure associated with those propagation instances.

A provisional structural relationship is:

Organizational Magnitude $\approx$ Organizational Multiplicity $\times$ Weaving Density

This relationship is not yet a mathematical definition of mass.

18.2 Hierarchical Mass

Mass is inherited across organizational levels while also receiving contributions from organization newly formed at each higher level.

First-Level Mass

First-Level mass originates in the Organizational Magnitude of the First-Level Organization's own Self-Closed Texture.

Second-Level Mass

Second-Level mass contains the organizational magnitude inherited from its First-Level constituents together with new organizational magnitude created through Interlocking Texture and Collective Closure.

Third-Level Mass

Third-Level mass contains inherited lower-level mass together with additional organizational contributions associated with higher-level collective organization and binding.

Therefore:

Higher-Level Mass = Inherited Organizational Magnitude + Newly Formed Organizational Magnitude

18.3 Mass and Collective Organization

Higher-level mass is not necessarily the arithmetic sum of constituent masses.

When organizations become mutually organized, the relationships among them become part of the organization of the whole and can contribute additional Organizational Magnitude.

The proton provides an important physical correspondence: its mass is dominated by collective QCD organization rather than by the small current masses of its valence quarks.

This supports the interpretation that higher-level organizational structure can contribute substantially to mass.

18.4 Mass and Large-Scale Organization

Organizational Magnitude can accumulate across increasingly higher levels of organization.

First-Level Organizations

↓

Second-Level Organizations

↓

Third-Level Organizations

↓

Higher-Level Matter

↓

Accumulated Organizational Magnitude

At large scales, the relationship between accumulated Organizational Magnitude and emergent spatial organization provides the basis for investigating gravity.

Gravity is not yet derived from Organizational Magnitude. Its relationship to emergent space-time remains an open research problem.

19. Time

19.1

Time is not a First Principle.

As the geometric existence of an organizational interaction, Open Texture itself possesses no time.

The temporal description associated with an observed organizational process emerges through Organizational Reading and Organizational Interpretation.

Therefore, time belongs to a higher-level description produced through Organizational Interpretation.

It is not a fundamental object of the Universe Crystal.

20. Space

20.1

Open Texture possesses:

- No length
- No distance
- No direction

A First-Level Organization established through Self-Closed Texture possesses identity and internal organization but does not possess independent spatial volume.

Spatiality emerges when multiple First-Level Organizations become mutually organized through Interlocking Texture and form a Collective Closure.

Collective Closure establishes a stable Organizational Topology in which relational distinctions corresponding to:

- Interior

- Boundary
- Exterior

can emerge.

Therefore:

Interlocking Organization → Collective Closure → Organizational Topology → Spatiality

Spatiality does not begin as a pre-existing geometric container. Relational topology precedes geometric description.

Three mutually interlocking First-Level Organizations are proposed as the minimum configuration capable of forming the relevant Collective Closure:

A → B → C → A

This three-member Collective Closure may provide the structural basis for three independent spatial degrees of freedom:

Three-Member Collective Closure → Three Spatial Degrees of Freedom → Three-Dimensional Space

A First-Level Organization can therefore acquire a spatial position through relational participation in emergent Organizational Space without possessing independent spatial volume.

Macroscopic three-dimensional space is interpreted as the large-scale manifestation of many mutually organized collective structures.

The mathematical derivation of spatial metric, distance, three spatial degrees of freedom, continuous spatiality, and the full correspondence between Collective Closure and three-dimensional space remains open research.

21. Entanglement

21.1 Entanglement and Coherence

Coherence is the state in which the closure phases of two organizational propagation closures remain synchronized.

Entanglement involves a deeper organizational relationship in which the propagation closure relationships themselves become coupled.

Therefore:

Coherence → Synchronized Closure Phase

Entanglement → Coupling of Propagation Closures

In coherence, the two organizations retain their individual propagation closures while maintaining a synchronized phase relationship.

In entanglement, the propagation closure relationships of the two organizations become coupled within a deeper organizational relationship.

21.2 Candidate Mechanism I — Coupling of Closure Shapes

The two organizations may retain their own propagation closures while the shapes of those closures become coupled.

Each organization preserves its individual closure and identity, while the closure structures become mutually coupled.

Such a coupling may persist when the organizations become spatially separated.

This is a candidate mechanism for entanglement.

A potential difficulty is that predetermined information encoded in coupled closure shapes could resemble a hidden-variable interpretation.

21.3 Candidate Mechanism II — Propagation Closure Across Two Particles

The propagation closure itself may extend across two particles, forming a common closure structure.

In this interpretation, the relational propagation structure includes both particles within a larger common organizational closure.

The particles may remain spatially separated while participating in the common organizational closure.

“Across” does not refer to an ordinary physical propagation process through spatial distance. It refers to a common relational propagation structure involving both organizations.

This mechanism is currently considered a more promising candidate for explaining entanglement, although its underlying structure remains unresolved.

21.4 Entanglement and Spatial Distance

Entanglement raises the possibility that spatial distance is not the fundamental measure of every organizational relationship.

Two organizations may remain spatially separated while participating in a common organizational relationship.

This suggests that some fundamental organizational relationships may exist independently of ordinary spatial distance.

The relationship between organizational connectivity and emergent spatial distance remains an open research question.